Introduction to Artificial intelligence

Tutorial Sheet 6 (Logic and Reasoning - Solution)

Exercise 1:

Consider a vocabulary with only four propositions, *A*, *B*, *C*, and *D*. How many interpretations (i.e. possible worlds) and how many models are there for the following sentences?

a. $B \vee C$.
b. $\neg A \vee \neg B \vee \neg C \vee \neg D$.
c. $(A \Rightarrow B) \wedge A \wedge \neg B \wedge C \wedge D$.
d. $(A \wedge B) \vee (C \wedge D)$.
e. $B \Rightarrow (A \wedge B)$.

Solution:

Since there are 4 variables in the vocabulary, there are 2^4 possible interpretations for each sentence.

A	B	C	D	Sent. a	Sent. b	Sent. c	Sent. d	Sent. e
T	T	T	T	T	F			
T	T	T	F	T	T			
T	T	F	T	T	T			
T	T	F	F	T	T			
T	F	T	T	T	T			
T	F	T	F	T	T			
T	F	F	T	F	T			
T	F	F	F	F	T			
F	T	T	T	T	T			
F	T	T	F	T	T			
F	T	F	T	T	T			
F	T	F	F	T	T			
F	F	T	T	T	T			
F	F	T	F	T	T			
F	F	F	T	F	T			
F	F	F	F	F	T			

The number of models is explained below:

- Note The error in answer (b): it should be: Sentence is false only if all are TRUE.
- For question (c) one may explain that (using distribution) the sentence is equivalent to:

$$(\neg A \wedge A \wedge \neg B \wedge C \wedge D) \vee (B \wedge A \wedge \neg B \wedge C \wedge D)$$

  Which is clearly False.
- For question (e) one can easily understand the answer by realizing that the sentence is equivalent to:

$$(\neg B \vee A) \wedge (\neg B \vee B)$$

These can be computed by counting the rows in a truth table that come out true, but each has some simple property that allows a short-cut:

a. Sentence is false only if B and C are false, which occurs in 4 cases for A and D, leaving 12.
b. Sentence is false only if A, B, C, and D are false, which occurs in 1 case, leaving 15.
c. The last four conjuncts specify a model in which the first conjunct is false, so 0.
d. 4 worlds satisfy $A \wedge B$, 4 satisfy $C \wedge D$, minus 1 for double-counting the model that satisfies both, leaving 7.
e. The sentence is true when B is false (8) and when B is true and A is true (4), so 12 in all.

Exercise 2:

Decide whether each of the following sentences is valid, unsatisfiable, or neither. Verify your decisions using truth tables or the equivalence rules of Figure 7.11 (page 223).

a. $Smoke \Rightarrow Smoke$

b. $Smoke \Rightarrow Fire$

c. $(Smoke \Rightarrow Fire) \Rightarrow (\neg Smoke \Rightarrow \neg Fire)$

d. $Smoke \vee Fire \vee \neg Fire$

e. $((Smoke \wedge Heat) \Rightarrow Fire) \Leftrightarrow ((Smoke \Rightarrow Fire) \vee (Heat \Rightarrow Fire))$

f. $(Smoke \Rightarrow Fire) \Rightarrow ((Smoke \wedge Heat) \Rightarrow Fire)$

g. $Big \vee Dumb \vee (Big \Rightarrow Dumb)$

$$
\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{De Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{De Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}
$$

Figure 7.11 Standard logical equivalences. The symbols α, β, and γ stand for arbitrary sentences of propositional logic.

Solution:

If a schema comes out true in one or more of its interpretation we called it satisfiable

Here i had checked using **Truth Table.**

so Method of checking is:-

1] A schema is valid if and only if its negation is unsatifiable. Here unsatifiable means always false.

2]A schema is unsatifiable if and only if its negation is valid.

3] If a schema comes out true in one or more of its interpretation we called it satisfiable

a) Here we we clearly see its negation in always false so this is a **valid statement** and also tautology is valid statement.

smoke	(smoke → smoke)
T	T
F	T

More precisely, the sentence $smoke \Rightarrow smoke$ is equivalent to $\neg\, smoke \vee smoke$ which is always true.

b)so according to rule 3 it is satisfiable and **neither** unsatisfiable nor valid.

Smoke	Fire	(smoke →Fire)
T	T	T
T	F	F
F	T	T
F	F	T

c) so according to rule 3 it is satisfiable and **neither** unsatisfiable nor valid.

Smoke	Fire	¬Smoke	¬Fire	(Smoke →Fire)	(¬Smoke→ ¬Fire)	((Smoke→Fire) → (¬Smoke→ ¬Fire))
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

d)Here we we clearly see its negation in always false so this is a **valid statement** and also tautology is valid statement.

Smoke	Fire	((Smoke ∨ Fire) ∨ ¬Fire)
T	T	T
T	F	T
F	T	T
F	F	T

One can see this by realizing that the sentence is equivalent to $smoke \vee (Fire \vee \neg Fire)$ $which\ is\ equivalent\ to$ $smoke \vee T$ i.e. to T

e)**valid statement**

P	Q	R	(P ∧ Q)	((P ∧ Q) → R)	(P → R)	((P → R) ∨ (Q → R))	(((P ∧ Q) → R) ↔ ((P → R) ∨ (Q → R)))
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	T	T	T
T	F	F	F	T	F	T	T
F	T	T	F	T	T	T	T
F	T	F	F	T	T	T	T
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

The validity of the sentence can also easily be shown by applying a sequence of equivalence rules to eventually reach T.

f)**valid statement**

Smoke	Fire	Heat	(Smoke → Fire)	(smoke ∧ Heat)	((Smoke → Fire) → ((smoke ∧ Heat) → Fire))
T	T	T	T	T	T
T	T	F	T	F	T
T	F	T	F	T	T
T	F	F	F	F	T
F	T	T	T	F	T
F	T	F	T	F	T
F	F	T	T	F	T
F	F	F	T	F	T

g)**valid statement**

Big	Dumb	(Big → Dumb)	((Big∨ Dumb∨ (Big → Dumb))
T	T	T	T
T	F	F	T
F	T	T	T
F	F	T	T

One can also prove the validity of sentences (f) and (g) by applying sequences of equivalences rules. For instance, for (f), it is as follows:

$$(S \rightarrow F) \rightarrow (S \wedge H \rightarrow F)$$

$$\neg (S \rightarrow F) \vee (S \wedge H \rightarrow F)$$

$$\neg (\neg S \vee F) \vee (\neg S \vee \neg H \vee F)$$

$$(S \wedge \neg F) \vee (\neg S \vee \neg H \vee F)$$

$$(S \vee \neg S \vee \neg H \vee F) \wedge (\neg F \neg S \vee \neg H \vee F)$$

$$T \wedge T$$

Proving the validity of (g) is even simpler

Exercise 3:

Which of the following are correct?

a. $False \models True$.
b. $True \models False$.
c. $(A \wedge B) \models (A \Leftrightarrow B)$.
d. $A \Leftrightarrow B \models A \vee B$.
e. $A \Leftrightarrow B \models \neg A \vee B$.
f. $(A \wedge B) \Rightarrow C \models (A \Rightarrow C) \vee (B \Rightarrow C)$.
g. $(C \vee (\neg A \wedge \neg B)) \equiv ((A \Rightarrow C) \wedge (B \Rightarrow C))$.
h. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B)$.
i. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B) \wedge (\neg D \vee E)$.
j. $(A \vee B) \wedge \neg(A \Rightarrow B)$ is satisfiable.
k. $(A \Leftrightarrow B) \wedge (\neg A \vee B)$ is satisfiable.
l. $(A \Leftrightarrow B) \Leftrightarrow C$ has the same number of models as $(A \Leftrightarrow B)$ for any fixed set of proposition symbols that includes A, B, C.

Solution:

In the following, each time be precise by defining the set of models of the LHS and the set of models of the RHS and showing whether M(LHS) ⊂ M(RHS).
Alternatively, one might prove the equivalence between the LHS and the RHS.

In all cases, the question can be resolved easily by referring to the definition of entailment.

a. $False \models True$ is true because $False$ has no models and hence entails every sentence AND because $True$ is true in all models and hence is entailed by every sentence.

b. $True \models False$ is false.

c. $(A \wedge B) \models (A \Leftrightarrow B)$ is true because the left-hand side has exactly one model that is one of the two models of the right-hand side.

d. $A \Leftrightarrow B \models A \vee B$ is false because one of the models of $A \Leftrightarrow B$ has both A and B false, which does not satisfy $A \vee B$.

e. $A \Leftrightarrow B \models \neg A \vee B$ is true because the RHS is $A \Rightarrow B$, one of the conjuncts in the definition of $A \Leftrightarrow B$.

f. $(A \wedge B) \Rightarrow C \models (A \Rightarrow C) \vee (B \Rightarrow C)$ is true because the RHS is false only when both disjuncts are false, i.e., when A and B are true and C is false, in which case the LHS is also false. This may seem counterintuitive, and would not hold if $\Rightarrow$ is interpreted as "causes."

g. $(C \vee (\neg A \wedge \neg B)) \equiv ((A \Rightarrow C) \wedge (B \Rightarrow C))$ is true; proof by truth table enumeration, or by application of distributivity (Fig 7.11).

h. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B)$ is true; removing a conjunct only allows more models.

i. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B) \wedge (\neg D \vee E)$ is false; removing a disjunct allows fewer models.

j. $(A \vee B) \wedge \neg(A \Rightarrow B)$ *is* satisfiable; model has A and $\neg B$.

k. $(A \Leftrightarrow B) \wedge (\neg A \vee B)$ *is* satisfiable; RHS is entailed by LHS so models are those of $A \Leftrightarrow B$.

l. $(A \Leftrightarrow B) \Leftrightarrow C$ does have the same number of models as $(A \Leftrightarrow B)$; half the models of $(A \Leftrightarrow B)$ satisfy $(A \Leftrightarrow B) \Leftrightarrow C$, as do half the non-models, and there are the same numbers of models and non-models.

For question (I) use a truth table count the number of models (4) for A ⇔B and the number of models (4) for A ⇔B ⇔ C

Exercise 4:

Consider a vocabulary with the following symbols:
Occupation(p, o): Predicate. Person p has occupation o.
Customer(p1, p2): Predicate. Person p1 is a customer of person p2.
Boss(p1, p2): Predicate. Person p1 is a boss of person p2.
Doctor, Surgeon, Lawyer, Actor: Constants denoting occupations.
Emily, Joe: Constants denoting people.
Use these symbols to write the following assertions in first-order logic:
a. Emily is either a surgeon or a lawyer.
b. Joe is an actor, but he also holds another job.
c. All surgeons are doctors.
d. Joe does not have a lawyer (i.e., is not a customer of any lawyer).
e. Emily has a boss who is a lawyer.
f. There exists a lawyer all of whose customers are doctors.
g. Every surgeon has a lawyer.

Solution:

In the following exercises, when checking the work of the students, please make sure they respect the syntax: propositions, function symbols, and predicate symbols all start with upper case letters, while variables start with lower case.

a. $O(E,S) \vee O(E,L)$.
b. $O(J,A) \wedge \exists p \;\; p \neq A \wedge O(J,p)$.
c. $\forall p \;\; O(p,S) \;\Rightarrow\; O(p,D)$.
d. $\neg\exists p \;\; C(J,p) \wedge O(p,L)$.
e. $\exists p \;\; B(p,E) \wedge O(p,L)$.
f. $\exists p \;\; O(p,L) \wedge \forall q \;\; C(q,p) \;\Rightarrow\; O(q,D)$.
g. $\forall p \;\; O(p,S) \;\Rightarrow\; \exists q \;\; O(q,L) \wedge C(p,q)$.

Exercise 5:

For each of the following sentences in English, decide if the accompanying first-order logic sentence is a good translation. If not, explain why not and correct it. (Some sentences may have more than one error!)

a. No two people have the same social security number.

$$\neg\exists x,y,n \;\; Person(x) \wedge Person(y) \;\Rightarrow\; [HasSS\#(x,n) \wedge HasSS\#(y,n)].$$

b. John's social security number is the same as Mary's.

$$\exists n \;\; HasSS\#(John,n) \wedge HasSS\#(Mary,n).$$

c. Everyone's social security number has nine digits.

$$\forall x,n \;\; Person(x) \;\Rightarrow\; [HasSS\#(x,n) \wedge Digits(n,9)].$$

d. Rewrite each of the above (uncorrected) sentences using a function symbol $SS\#$ instead of the predicate $HasSS\#$.

Solution:

a. "No two people have the same social security number."

$$\neg\exists\, x, y, n \;\; Person(x) \wedge Person(y) \;\Rightarrow\; [HasSS\#(x, n) \wedge HasSS\#(y, n)].$$

This uses $\Rightarrow$ with $\exists$. It also says that no person has a social security number because it doesn't restrict itself to the cases where x and y are not equal. Correct version:

$$\neg\exists\, x, y, n \;\; Person(x) \wedge Person(y) \wedge \neg(x = y) \wedge [HasSS\#(x, n) \wedge HasSS\#(y, n)]$$

b. "John's social security number is the same as Mary's."

$$\exists\, n \;\; HasSS\#(John, n) \wedge HasSS\#(Mary, n).$$

This is OK.

c. "Everyone's social security number has nine digits."

$$\forall\, x, n \;\; Person(x) \;\Rightarrow\; [HasSS\#(x, n) \wedge Digits(n, 9)].$$

This says that everyone has every number. $HasSS\#(x, n)$ should be in the premise:

$$\forall\, x, n \;\; Person(x) \wedge HasSS\#(x, n) \;\Rightarrow\; Digits(n, 9)$$

d. Here $SS\#(x)$ denotes the social security number of x. Using a function enforces the rule that everyone has just one.

$$\neg\exists\, x, y \;\; Person(x) \wedge Person(y) \;\Rightarrow\; [SS\#(x) = SS\#(y)]$$
$$SS\#(John) = SS\#(Mary)$$
$$\forall\, x \;\; Person(x) \;\Rightarrow\; Digits(SS\#(x), 9)$$

Exercise 6:

A. Rewrite the following statements in First-Order Logic. Choose appropriate predicates and constants.

1. All dogs howl at night.
2. Anyone who has any cats will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. Tariq has either a cat or a dog.
5. (Conclusion) If Tariq is a light sleeper, then Tariq does not have any mice.

B. Convert the statements in CNF showing all the steps.

C. Apply the Resolution inference rule to prove the conclusion (statement 5) from the first four statements (statements 1 to 4). Make sure that each step all the substitutions are shown as well as the parent clauses and the resolvent.

Solutions:

A.

1. *∀x (DOG(x) → HOWL(x, Night))*
2. *∀x ∀y (HAVE (x,y) ∧ CAT (y) → ¬ ∃z (HAVE(x,z) ∧ MOUSE (z)))*
3. *∀x (LS(x) → ¬ ∃y (HAVE (x,y) ∧ HOWL(y, Night))*
4. *∃x (HAVE (Tariq,x) ∧ (CAT(x) ∨ DOG(x)))*
5. *LS(Tariq) → ¬ ∃z (HAVE(Tariq,z) ∧ MOUSE(z))*

B.

1. *∀x (DOG(x) → HOWL(x, Night))*
   ¬ DOG(x) ∨ HOWL(x, Night)
2. *∀x ∀y (HAVE (x,y) ∧ CAT (y) → ¬ ∃z (HAVE(x,z) ∧ MOUSE (z)))*

$\forall x\ \forall y\ (HAVE\ (x,y) \wedge CAT\ (y) \rightarrow \forall z \neg\ (HAVE(x,z) \wedge MOUSE\ (z)))$

$\forall x\ \forall y\ \forall z\ (\neg\ (HAVE\ (x,y) \wedge CAT\ (y)) \vee \neg\ (HAVE(x,z) \wedge MOUSE\ (z)))$

$\neg\ HAVE(x,y) \vee \neg\ CAT(y) \vee \neg\ HAVE(x,z) \vee \neg\ MOUSE(z)$

3. $\forall x\ (LS(x) \rightarrow \neg\ \exists y\ (HAVE\ (x,y) \wedge HOWL(y, Night))$
   $\forall x\ (LS(x) \rightarrow \forall y \neg\ (HAVE\ (x,y) \wedge HOWL(y, Night))$

   $\forall x\ \forall y\ (LS(x) \rightarrow \neg\ HAVE(x,y) \vee \neg\ HOWL(y, Night))$

   $\forall x\ \forall y\ (\neg\ LS(x) \vee \neg\ HAVE(x,y) \vee \neg\ HOWL(y, Night))$

   $\neg\ LS(x) \vee \neg\ HAVE(x,y) \vee \neg\ HOWL(y, Night)$

4. $\exists x\ (HAVE\ (Tariq,x) \wedge (CAT(x) \vee DOG(x)))$
   $HAVE(Tariq,A) \wedge (CAT(A) \vee DOG(A))$

5. $\neg\ [LS(Tariq) \rightarrow \neg\ \exists z\ (HAVE(Tariq,z) \wedge MOUSE(z))]$ (negated conclusion)
   $\neg\ [\neg\ LS\ (Tariq) \vee \neg\ \exists z\ (HAVE\ (Tariq, z) \wedge MOUSE(z))]$

   $LS(Tariq) \wedge \exists z\ (HAVE(Tariq, z) \wedge MOUSE(z)))$

   $LS(Tariq) \wedge HAVE(Tariq,B) \wedge MOUSE(B)$

   Or you may give the converted sentence here and negate it below, yielding the 3 clauses under 5 below.

The set of CNF clauses for this problem is thus as follows:

1. $\neg\ DOG(x) \vee HOWL(x, Night)$
2. $\neg\ HAVE(x,y) \vee \neg\ CAT(y) \vee \neg\ HAVE(x,z) \vee \neg\ MOUSE(z)$
3. $\neg\ LS(x) \vee \neg\ HAVE(x,y) \vee \neg\ HOWL(y, Night)$
4.
   a. $HAVE(Tariq,A)$
   b. $CAT(A) \vee DOG(A)$
5.
   a. $LS(Tariq)$
   b. $HAVE(Tariq,B)$
   c. $MOUSE(B)$

C.

The resolution-based proof tree is being presented under a tabular format (just for convenience of not having to draw a tree):

Parent clause numbers	**Resolvent Number**	***Resolvent***	***Applied Substitution***
[1 4.b]	6.	$CAT(A) \vee HOWL(A, Night)$	$\{x/A\}$
[2 5.c]	7.	$\neg\ HAVE(x,y) \vee \neg\ CAT(y) \vee \neg\ HAVE(x,B)$	$\{z/B\}$

[7 5.b]	8.	¬ *HAVE(Tariq,y)* ∨ ¬ *CAT(y)*	*{x/Tariq}*
[6 8]	9.	¬ *HAVE(Tariq,A)* ∨ *HOWL(A, Night)*	*{y/A}*
[4.a 9]	10.	*HOWL(A, Night)*	*{}*
[3 10]	11.	¬ *LS(x)* ∨ ¬ *HAVE(x,A)*	*{y/A}*
[4.a 11]	12.	¬ *LS(Tariq)*	*{x/Tariq}*
[5.a 12]	13.	□	*{}*

Empty clause reached, QED.